

Solving the 1D Wave Equation

Tuesday, July 8, 2025 9:09 AM

2. Solving the 1D Wave Equation using Fourier Series

2.1 The one-dimensional wave equation is:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

where c is a constant (speed of the wave).

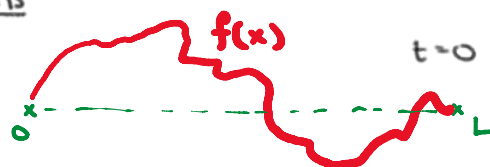
2.2 We will assume boundary conditions

$$u(0, t) = 0 \quad \text{and} \quad u(L, t) = 0 \quad \text{for all } t \geq 0$$

2.3 We will also assume initial conditions

initial deflection $u(x, 0) = f(x)$

initial velocity $u_t(x, 0) = g(x)$ i.e. $\left. \frac{\partial u(x, t)}{\partial t} \right|_{t=0} = g(x)$



2.4 Our first attempt will assume a "separation of variables"

$$\text{Let: } u(x, t) = F(x) G(t)$$

In this case: $\frac{\partial u}{\partial x} = F'(x)G(t)$ and $\frac{\partial u}{\partial t} = F(x)G'(t)$

Thus: $\frac{\partial^2 u}{\partial x^2} = F''(x)G(t)$ and $\frac{\partial^2 u}{\partial t^2} = F(x)G''(t)$

So that the wave equation: $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

becomes: $F(x)G''(t) = c^2 F''(x)G(t)$

Separate x & t : $\frac{1}{c^2} \frac{G''(t)}{G(t)} = \frac{F''(x)}{F(x)}$
depends only on t depends only on x

Each side must be constant, so: $\frac{1}{c^2} \frac{G''(t)}{G(t)} = k$, $\frac{F''(x)}{F(x)} = k$

Thus, we get two ODEs:

$$G''(t) - kc^2 G(t) = 0 \quad \text{and} \quad F''(x) - kF(x) = 0$$

We know how to (generally) solve these:

e.g. $F''(x) - kF(x) = 0 \Rightarrow F(x) = A e^{\sqrt{k}x} + B e^{-\sqrt{k}x}$ if $k > 0$
or $F(x) = A \cos(\omega x) + B \sin(\omega x)$ where $\omega^2 = -k$ if $k < 0$
or $F(x) = Ax + B$ if $k = 0$

2.5 Instead of solving right away, consider the boundary conditions to see if there is more information:

$$u(0, t) = F(0)G(t) = 0 \quad \text{and} \quad u(L, t) = F(L)G(t) = 0$$

Consider $G(t) \neq 0$, then $F(0) = 0 = F(L)$

Now, for $k \geq 0$, this would result in $A = B = 0$, meaning that as a function $F = 0$ (i.e., $F(x)$ is zero everywhere)

We want $F \neq 0$, which leaves us with only one option: $k < 0$
and thus $k = -\omega^2$, so that

$$F(x) = A \cos(\omega x) + B \sin(\omega x)$$

$$\text{Now: } F(0) = 0 \Rightarrow A \cos(0) + B \sin(0) = 0 \Rightarrow A = 0$$

$$\text{Thus } F(x) = B \sin(\omega x)$$

$$\text{Finally: } F(L) = 0 \Rightarrow B \sin(\omega L) = 0$$

we want $B \neq 0$, which means $\sin(\omega L) = 0$

$$\text{Thus: } \omega L = n\pi \quad \text{for integer value of } n \\ \Rightarrow \omega = \frac{n\pi}{L}$$

$$\text{Hence } F(x) = B \sin\left(\frac{n\pi x}{L}\right)$$

2.6 As $\omega = \frac{n\pi}{L}$ and $k = -\omega^2$, then $k = -\left(\frac{n\pi}{L}\right)^2$.

$$\text{Recall: } G''(t) - kc^2 G(t) = 0$$

Recall: $G''(t) - kc^2 G(t) = 0$

$$\Rightarrow G''(t) + \left(\frac{n\pi c}{L}\right)^2 G(t) = 0$$

Let $\lambda_n = \frac{n\pi c}{L} \Rightarrow G''(t) + \lambda_n^2 G(t) = 0$

$$\Rightarrow G(t) = P \cos(\lambda_n t) + Q \sin(\lambda_n t)$$

So for, we have:

$$\begin{aligned} u(x, t) &= F(x)G(t) = B \sin\left(\underbrace{\frac{n\pi}{L} x}_{\lambda_n/c}\right) [P \cos(\lambda_n t) + Q \sin(\lambda_n t)] \\ &= \left[\underbrace{BP}_{A_n} \cos(\lambda_n t) + \underbrace{BQ}_{B_n} \sin(\lambda_n t) \right] \sin\left(\frac{n\pi}{L} x\right) \\ &= [A_n \cos(\lambda_n t) + B_n \sin(\lambda_n t)] \sin\left(\frac{n\pi}{L} x\right) \end{aligned}$$

2.3 Can we directly use initial conditions?

The short answer is: No!

e.g. if we try $u(x, 0) = f(x)$

$$\Rightarrow [A_n \cos(0) + B_n \sin(0)] \sin\left(\frac{n\pi}{L} x\right) = f(x)$$

$$\Rightarrow A_n \sin\left(\frac{n\pi}{L} x\right) = f(x)$$

But what if $f(x)$ is not a sine function? we have no control on $f(x)$ or $g(x)$

This is where Fourier Series comes to play

Q: What do you remember about Fourier Series?

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(u_n x) + b_n \sin(u_n x))$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(u_n x) + b_n \sin(u_n x))$$

enforcing $f(0)=0$ and $f(L)=0$, we may write

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right)$$

where the Fourier coefficients: $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$

Q: What is the relevance?

THEOREM 1

Fundamental Theorem on Superposition

If u_1 and u_2 are solutions of a homogeneous linear PDE in some region R , then

$$u = c_1 u_1 + c_2 u_2$$

with any constants c_1 and c_2 is also a solution of that PDE in the region R .

Instead of assuming $u(x,t) = F(x)G(t)$,

we assume: $u(x,t) = \sum_{n=1}^{\infty} F_n(x)G_n(t)$

In other words, insist:

$$u(x,t) = \sum_{n=1}^{\infty} [A_n \cos(\lambda_n t) + B_n \sin(\lambda_n t)] \sin\left(\frac{n\pi}{L} x\right)$$

Now if we use $u(x,0) = f(x)$, then

$$\sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L} x\right) = f(x) \text{ makes sense as a Fourier series with}$$

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

Similarly:

$$u_t(x,t) = \sum_{n=1}^{\infty} [-A_n \lambda_n \sin(\lambda_n t) + B_n \lambda_n \cos(\lambda_n t)] \sin\left(\frac{n\pi}{L} x\right)$$

so setting $u_t(x,0) = g(x)$

so setting $u_t(x, 0) = g(x)$

$$\Rightarrow \sum_{n=1}^{\infty} B_n \lambda_n \sin\left(\frac{n\pi}{L} x\right) = g(x) \quad \text{as a Fourier series,}$$

and thus

$$B_n = \frac{1}{\lambda_n} \left(\frac{2}{L}\right) \int_0^L g(x) \sin\left(\frac{n\pi}{L} x\right) dx \quad \left(\text{Recall: } \lambda_n = \frac{n\pi c}{L}\right)$$

$$\Rightarrow B_n = \frac{2}{c n \pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

Remark: we never established that $g(x)$ can be written exclusively as a Fourier sine series: a detail that would not be addressed in this course.

2.8

EXAMPLE 1 Vibrating String if the Initial Deflection Is Triangular

Find the solution of the wave equation (1) satisfying (2) and corresponding to the triangular initial deflection

$$f(x) = \begin{cases} \frac{2k}{L}x & \text{if } 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & \text{if } \frac{L}{2} < x < L \end{cases}$$

and initial velocity zero. (Figure 291 shows $f(x) = u(x, 0)$ at the top.)

Solution. Since $g(x) = 0$, we have $B_n^* = 0$ in (12), and from Example 4 in Sec. 11.3 we see that the B_n are given by (5), Sec. 11.3. Thus (12) takes the form

$$u(x, t) = \frac{8k}{\pi^2} \left[\frac{1}{1^2} \sin \frac{\pi}{L} x \cos \frac{\pi c}{L} t - \frac{1}{3^2} \sin \frac{3\pi}{L} x \cos \frac{3\pi c}{L} t + \dots \right].$$